SIMATS ENGINEERING

CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA1522

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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 50**

Question Count: 5

Code of Conduct

I **Dinnepati Sindhu Prasad** (192311271) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Dinnepati Sindhu Prasad

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	**Weightage**	**Level 4 - Exemplary**	**Level 3 – Proficient**	**Level 2 - Developing**	**Level 1 - Beginning**
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear causeeffect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

Answers

1.Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting missioncritical applications. Analyze performance, security, and manageability, then justify the better choice.

Sol: **Type-1 Hypervisor (Bare-Metal):**

- Installed directly on the physical hardware.
- Does not require a host operating system.
- Examples: VMware ESXi, Microsoft Hyper-V, Citrix Hypervisor.

Type-2 Hypervisor (Hosted):

- Runs on top of a host operating system.
- Suitable for desktops, testing, and development.
- Examples: Oracle VM VirtualBox and VMware Workstation

Feature	Type-1 Hypervisor	Type-2 Hypervisor
Installation	Directly on hardware	On host operating system
Performance	Very high	Moderate
Security	Strong	Lower due to host OS dependency
Resource Overhead	Low	High
Reliability	Excellent	Moderate
Management	Centralized	Local management
Best Use	Enterprise cloud datacenters	Personal use and testing

Performance Analysis

- Type-1 hypervisors access hardware directly, reducing CPU and memory overhead.

- Type-2 hypervisors depend on the host operating system, increasing latency and reducing efficiency.

Security Analysis

- Type-1 provides better isolation between virtual machines.
- Smaller attack surface because there is no host OS.
- Type-2 is vulnerable if the host operating system is compromised.

Manageability:

- Type-1 supports centralized management, live migration, clustering, and high availability.
- Type-2 mainly supports individual systems with limited enterprise features.

Justification:

For mission-critical cloud applications, Type-1 Hypervisor is the better choice because it offers:

- Higher performance
- Better security
- Improved scalability
- High availability
- Enterprise-grade management

2.A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Sol) Workload Isolation Using Virtualization Architecture

A cloud provider hosts Finance, Healthcare, and Student Portal applications on the same physical server. Strong isolation is required to protect sensitive information.

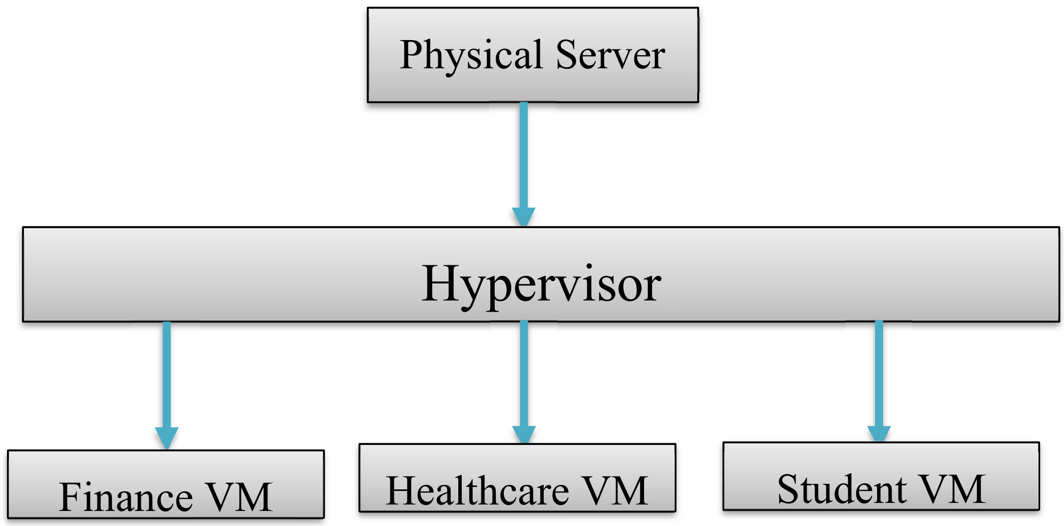
How Virtualization Supports Isolation:

1. Each application runs inside its own Virtual Machine (VM).
2. Hypervisor allocates dedicated CPU, memory, and storage resources.
3. Virtual networking separates traffic using VLANs or virtual switches.
4. Access control policies restrict communication between workloads.
5. Snapshots and VM monitoring improve recovery and auditing.

Isolation Benefits:

- Prevents unauthorized data access.
- Stops failures in one VM from affecting others. • Improves compliance with security regulations.
- Enables secure resource sharing.

Isolation Architecture:



Major Attack Surfaces:

1. Hypervisor Attack

- Attackers exploit vulnerabilities in the hypervisor.
- Can gain control over multiple virtual machines.
- **Mitigation:**
  - Regular security updates
  - Secure boot
  - Strong administrator authentication

2. VM Escape Attack:

- Malware escapes from one VM into the hypervisor or another VM.
- Can compromise multiple workloads.
- **Mitigation:**

- Patch hypervisors regularly
- Use strong VM isolation
- Monitor abnormal VM behaviour

Conclusion:

Virtualization provides strong isolation through independent virtual machines and resource separation. Securing the hypervisor and preventing VM escape attacks are essential for protecting sensitive workloads.

3)Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Sol) VM Host Utilization Analysis

Given Data

- CPU Utilization = 92%
- Memory Utilization = 88%
- Storage I/O Latency = 35 ms
- Average VM Boot Time = 170 seconds

Analysis

CPU Bottleneck:

- CPU usage is extremely high.
- Virtual machines compete for processor resources.
- Effects:
    - Slow application response
    - Increased VM scheduling delays

Memory Bottleneck:

- Memory usage is close to maximum.
- Hypervisor may start memory swapping.
- **Effects:**
    - Performance degradation
    - Longer VM startup time

Storage Bottleneck:

- Storage latency of 35 ms is much higher than the recommended value (typically below 10 ms).

- Disk operations become slow.
- **Effects:**
    - Delayed VM boot
    - Slow database performance
    - Slow file access

Boot Time Analysis:

- Normal VM boot time is usually less than one minute.
- 170 seconds indicates severe storage and resource contention.

Corrective Actions:

1. Add additional CPU resources or migrate some VMs.
2. Increase physical RAM.
3. Upgrade storage to SSD/NVMe.
4. Balance workloads across multiple hosts.
5. Enable resource monitoring and load balancing.
6. Remove unused VMs and optimize resource allocation.

Conclusion:

The major bottlenecks are:

- CPU saturation
- Memory pressure
- High storage latency
- Upgrading hardware and redistributing workloads will significantly improve VM performance.

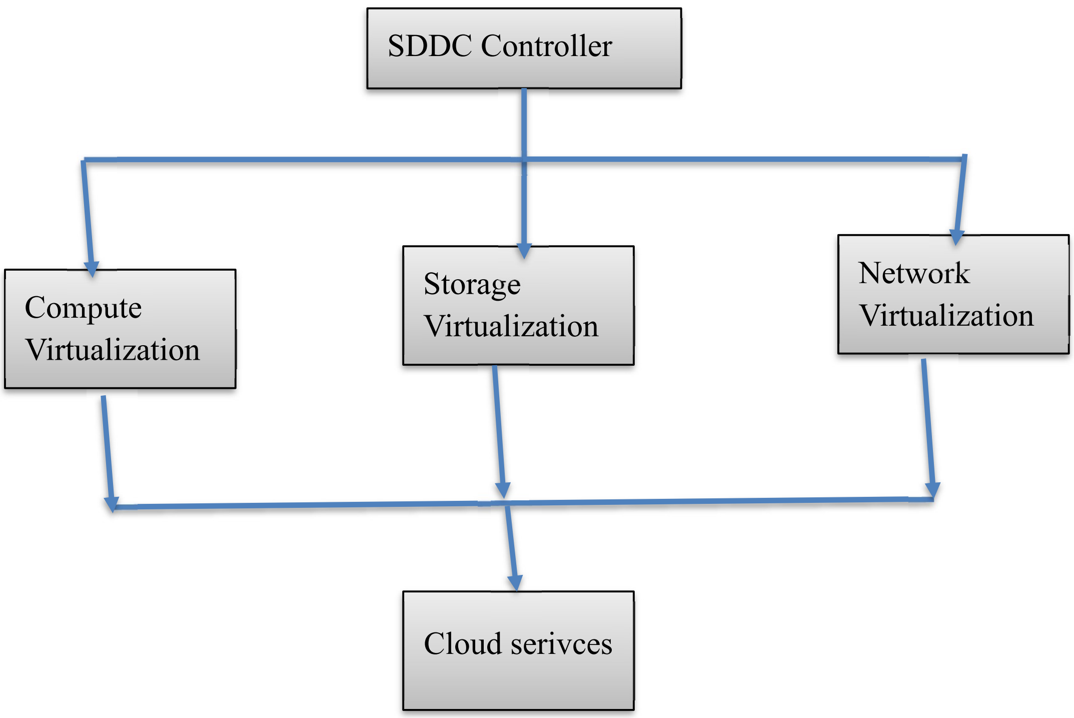
4.Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Sol) A Software Defined Datacenter (SDDC) virtualizes all infrastructure resources and manages them through software, unlike a traditional datacenter that relies on dedicated physical hardware.

Feature	Traditional Datacenter	Software Defined Datacenter
Compute	1. Physical servers	1. Virtual machines

Storage	2. Dedicated storage devices	2. Software-defined storage
Network	3. Physical switches	3. Software-defined networking
Provisioning	4. Manual	4. Automated
Scalability	5. Limited	5. Highly scalable
Management	6. Hardware-based	Software-controlled

Software Defined Datacenter:



Compute Virtualization:

- Multiple virtual machines share physical servers.
- Resources can be allocated dynamically.
- Live migration minimizes downtime.

Storage Virtualization:

- Combines storage from multiple devices into a single virtual storage pool.
- Improves utilization and simplifies management.
- Supports snapshots and replication.

Network Virtualization:

- Creates virtual networks independent of physical hardware.
- Enables rapid network provisioning.
- Improves security using virtual firewalls and micro-segmentation.

How SDDC Improves Agility:

- Rapid deployment of virtual resources.
- Automatic scaling according to workload.
- Centralized management.
- Faster disaster recovery.
- Reduced operational cost.
- Better resource utilization.

Conclusion:

Software Defined Datacenters improve agility through automated compute, storage, and network virtualization, making cloud infrastructure more flexible, scalable, and efficient than traditional datacenters.

5.) A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Sol) Identity Federation in Multi-Cloud
Problem Analysis:

A multi-cloud federation project fails because each cloud provider maintains separate user identities and authentication systems.

Issues:

- Different usernames across providers.
- Duplicate user accounts.
- Inconsistent access permissions.
- Poor user experience.
- Increased security risks.

Secure Identity Federation Design:

1. Central Identity Provider (IdP)

- Users authenticate once using a trusted Identity Provider.
- Authentication is shared across all cloud providers.

2. Single Sign-On (SSO):

- Users log in once.
- Access multiple cloud services without repeated authentication.

3. Federation Standards

Use industry standards such as:

- SAML (Security Assertion Markup Language)
- OAuth 2.0
- OpenID Connect (OIDC)

4. Privacy Protection:

- Encrypt identity information during transmission.
- Share only the minimum required user attributes.
- Apply role-based access control (RBAC).
- Maintain audit logs for accountability.

5. Multi-Factor Authentication (MFA):

- Combine passwords with OTPs or biometric authentication.
- Protect against credential theft.

Benefits

- Secure authentication across clouds.
- Single Sign-On improves user experience.
- Reduced identity duplication.
- Strong privacy protection.
- Easier centralized management.

Conclusion

A centralized Identity Provider with SSO, federation standards (SAML, OAuth 2.0, and OpenID Connect), encryption, RBAC, and MFA provides secure identity management and privacy protection for multi-cloud environments.